How to write (and review) a Genome Report

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Abstract

As eukaryotic genomes continue to be sequenced, assembled, and annotated at decreasing cost and increasing rate, it is important that basic notification of the availability of these genomes, along with essential information about them, be made available to the community. One way of doing this is through publication of a "genome report", a short article format that several journals have established. Here, we provide a brief guide to the genome report and its main components, with recommendations for authors, reviewers, and publishers of such reports.

Keywords: genome resources, genome note, i5k initiative, genome sequencing, genome assembly

Significance

As more journals begin to publish specific "genome report" articles, authors and peer reviewers need to understand the basic requirements for such a format. This article provides guidelines and recommendations, along with a helpful checklist.

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Introduction

Eukaryotic genomes continue to be sequenced at an unprecedented pace, with over 10,000 submitted to INSDC databases in 2024 alone. As whole-genome sequencing becomes an almost commonplace scientific endeavor, findings that were once the subjects of detailed publication in the highest impact journals are now often summarized in a brief genome "note" or "report" in a variety of publication venues. However, these brief reports are vital to disseminate notice of newly-available genomes; provide essential metrics, statistics, and assorted metadata about these genomes; and ensure credit to the dedicated scientists whose time and effort was spent in sample collection, genome sequencing, assembly, and annotation. In this brief guide, we describe the basic components of reports of eukaryotic genomes and make recommendations for consideration by authors, peer reviewers, and publishers of these reports.

What minimum information should be in my genome report?

The content of your genome report will depend on whether the goal is to present a brief report of a newly sequenced genome or an in-depth paper with thorough analysis of a new genome and its contents, comparisons to previously published genomes, and significant new biological insights. The latter would typically be considered a traditional research paper, and should follow the relevant journal guidelines. Nevertheless, it should still include the essential information that would be contained in a brief, stand-alone report. Although the criteria vary from journal to journal, there are many commonalities as to what

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minimal information should be present in a genome report. A number of journals, including *Genome Biology and Evolution (GBE)*, have created dedicated article types with specific formats, expectations, and review criteria. In this section we highlight the most common required items, based on an evaluation of the guidelines for ten such journals (**Table 1**) and our previous assessment of best practices for genome sequencing projects (Tvedte, et al. 2026). Although few journals require all of them (**Table 2**), we suggest that a complete genome report should address each of the listed items. Indeed, many reporting practices have been widely adopted despite not being explicitly required or recommended (**Table 2**, final column; see (Tvedte, et al. 2026)). However, the reader should keep in mind that individual journals may also have other or additional requirements.

Although perhaps obvious, it is worth noting at the outset that a genome report is still a scientific paper and as such is reporting results that should be based on sound methodology, be backed by clear evidence, and have well-supported conclusions. Authors, reviewers, and readers alike need to bear this in mind and approach a genome report with the same critical evaluation of its claims that they would any other publication.

Any genome report should contain a certain set of minimal information. This can be divided into four basic components: metadata, methods, metrics, and data. **Table 2** lists the basic recommended information for each of these categories. *Metadata* refers to the "data about the data," and conveys details about how and where sequenced specimens were procured, the sex of the sequenced individual(s), what tissue(s) was sequenced, etc. NCBI provides an extensive set of BioSample "packages" which list both required and

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optional attributes for defining different types of BioSamples (https://www.ncbi.nlm.nih.gov/biosample/docs/packages/). These templates provide a good starting source for metadata needs. A minimal set of required metadata fields is described in the FAIR-bioHeaders Reference genome (FHR) specification (Wright, et al. 2024). A noteworthy aspect of this specification is that it enables the metadata to be tied to a reference genome as a component of the sequence's FASTA header, thereby keeping the genome sequence and its provenance metadata linked. A more extensive proposed standard for genome sequence metadata (MIGS: Minimum Information about a Genome Sequence) has been developed by the Genomic Standards Consortium and includes a checklist of suggested metadata (Field, et al. 2008). Metadata descriptions should be as specific as possible—which typically means including more metadata than required by the template—and provided with an eye toward future re-use of the data by other researchers. In general, the more comprehensive the metadata, the greater the likelihood that your data can and will be reused by others for a variety of purposes.

Methods should describe those used for DNA extraction, library preparation, and sequencing; assembly methods; assembly curation including contamination removal (Howe, et al. 2021); annotation strategy; and any other methods used to generate the data. All manual curation decisions should be described in the text or in a linked and accessible data repository (for example in AGP or assembly review files generated by curation incorporation software, or command logs from manual sequence rearrangements). Protocols should be well-referenced and any custom scripts and pipelines should be documented in detail and made available without restriction, ideally as a workflow for

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better reproducibility (Tvedte, et al. 2026). It is particularly important to note any deviations from standard protocols, especially if these were performed for specific troubleshooting reasons, e.g., if modifications to typical DNA extraction protocols were necessary in order to obtain high-quality genomic DNA. Not only is it important that such changes be well documented, but this information may be useful to researchers interested in sequencing other genomes from within the same or similar taxonomic groups.

Metrics about an assembly are frequently best presented as a table and/or figure (for example, *GBE* requires a concise summary of assembly metrics in table format). Requirements for which metrics must be included vary among the journals we surveyed (**Table 2**; also see **Glossary** in **Supplemental File 1**). However, many journals require a description of the assembled genome size, the number of contigs and contig N50, number of scaffolds and scaffold N50 (if applicable), and an assessment of genome completeness (most of the surveyed journals specifically cite BUSCO for this assessment, but we assume that other methods—e.g. OMArk (Nevers, et al. 2025), compleasm (Huang and Li 2023)—would be acceptable to most journals as suitable alternatives). When reporting BUSCO or similar completeness analyses, authors should provide the full result summary, including the lineage dataset and software version used and the proportions of complete single-copy, complete duplicated, fragmented, and missing markers. Reporting only the combined completeness value can obscure important differences among assemblies, since duplicated markers may reflect biological signal, such as polyploidy or recent gene duplication, or technical issues such as uncollapsed haplotypes.

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We also strongly recommend adding statistics of coverage by sequence in contigs, scaffolds, or chromosomes, base accuracy (QV), k-mer completeness values, and an analysis of repeats including transposable elements and telomeric repeats (particularly if the assembly is presented as chromosome-scale). Each of these metrics conveys a distinct aspect of how well the assembly is supported by the underlying sequencing data (QV measures if the assembly sequence is observed within the input reads, k-mer completeness measures the fraction of input read sequence captured by the assembly) and how well the assembly reflects biological reality (are sequences contiguous to chromosome-scale, do counts of chromosome-scale sequences match the observed karyotype, are expected genes—e.g. the BUSCO lineage benchmark set—fully represented). Reporting these metrics together in a compact format helps other researchers evaluate the strengths and weaknesses of the assembly and provides confidence in its integrity. Tools such as PAQman can assess many of these metrics in one workflow, facilitating assembly of a comprehensive report (O'Donnell, et al. 2026). **Box 1** provides links to both standards and software to aid in generating the recommended metrics.

Common figure formats for relevant assembly metrics are snail plots and blob plots that combine the distributions of sequence length, read coverage, and GC content with assembly metrics like N50 and BUSCO completeness (Challis and Blaxter 2026). Tables and figures are easily generated using the BlobToolKit software package (Challis, et al. 2020). An example set of plots is shown in **Figure 1**, and a detailed introduction to the snail plot is provided by Challis and Baxter (Challis and Blaxter 2026).

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For assemblies presented as chromosome-scale, near-complete, or telomere-to-telomere, authors should also report evidence supporting those claims, where available. This may include the expected haploid chromosome number or known karyotype for the species or a close relative, the number and proportion of scaffolds assigned to chromosomes or pseudomolecules, and the presence, number, and position of telomeric repeat motifs at scaffold ends. (Relevant methods can be found via the resources provided in **Box 1**.) However, these values should be interpreted cautiously, as expected chromosome numbers and telomeric repeat motifs may be unknown, variable, or non-canonical in some taxa.

Data–used here to refer to the raw sequence data, assembled genome sequence, and annotation itself–should be freely available and adhere to FAIR (Findable, Accessible, Interoperable, Reusable) principles (Wilkinson, et al. 2016). It is imperative that all data are deposited in archives that provide a persistent identifier, so that data remain findable and accessible well into the future. Data should conform to format standards, e.g. FASTA for sequence and GFF3 or GTF for annotations. Both raw sequence data and assembled genome data should be deposited in an appropriate INSDC repository (Karsch-Mizrachi, et al. 2025) and accession numbers should be provided. Annotation data should also be deposited in an appropriate repository, ideally an INSDC repository (for example, *GBE* requires that protein annotations are available in an INSDC repository). However, as the somewhat loose requirements of the GFF3 specification can make submission of annotation data to these repositories challenging, data generators should at a minimum submit annotation data to a generalist repository, e.g. Dryad (Dryad Digital Data

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Repository), that can provide a Digital Object Identifier (DOI) and be linked in the report. The new public version of the NCBI Eukaryotic Genome Annotation Pipeline (EGAPx; https://github.com/ncbi/egapx) generates submission-ready annotations, and as such may help lower the barrier to INSDC annotation submission.

What additional information should be in my genome report?

Additional information will depend on what methods were used in generating the genome assembly and what additional analyses were carried out. Recommended items include a description of sex chromosomes (where relevant), contact map data from Hi-C or similar scaffolding methods, identification and separate assembly/annotation of organelles (e.g., mitochondria), and haplotype information generated by phased assembly methods (for links to useful assembly tools and workflows, see **Box 1**). As always, a full description of the methods should be provided, including all software and parameters used and a complete account of manual curation decisions. Suggested additional annotation data include protein-coding annotation, BUSCO analysis performed in transcriptome or protein mode (after filtering for single isoforms per gene), other structural features such as ncRNAs, and functional annotations using GO and similar resources (for links to useful annotation tools, see **Box 1**). Note that some of these analyses are required by individual journals, and the specific journal requirements should be checked before submission.

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Although only required by some of the journals we surveyed, we also recommend that a genome report include evidence demonstrating that the sequenced species was properly identified as the correct species—preferably with links to appropriate vouchering information—as well as a statement verifying that the source sample was obtained both ethically (e.g., in accordance with the Nagoya Protocol of the Convention on Biological Diversity (Convention on Biological Diversity 2010)) and legally based on local law (see recommendations in (Lawniczak, et al. 2025)). For instance, *GBE* explicitly requires evidence that the Nagoya protocol was followed in cases where samples were taken from a country other than the author's.

While falling outside the realm of minimally required information, we also encourage authors to carefully consider and add any additional aspects of their methodology or results that would facilitate re-use of the genome, or be of assistance to those working with other members of the same or a similar taxonomic group (e.g., changes to standard DNA extraction protocols, taxon-specific repeat databases, sequence/chromosome naming conventions and homology with related species, etc.). A checklist of recommended data, metadata, metrics, and methods descriptions can be found in **Supplemental File 2**.

Where can I find standards/best practices?

Genome sequencing and assembly methods are continually evolving, and it can be difficult to keep up with current best practices and recommended standards. The Earth BioGenome

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Project (EBP)(Lewin, et al. 2022) has developed many such standards and practices, including recommended software tools, which can be found at the EBP website (https://www.earthbiogenome.org). These are also summarized in a user-friendly format by the European Reference Genome Atlas consortium (https://github.com/ERGA-consortium/EARs/wiki/About-the-quality-standard). As mentioned above, metadata standards have been developed by the Genomic Standards Consortium and promulgated by the NCBI BioSample (https://www.ncbi.nlm.nih.gov/biosample/docs/packages/) and EBI BioSamples (Courtot, et al. 2019) repositories. The i5k initiative (i5k Consortium 2013), an international organization that aims to enable access to high-quality genome data and metadata from insects and other arthropods, has reviewed both existing standards and levels of adherence to these standards, and recommended places where there is a need for increased awareness and/or development of new standards (Tvedte, et al. 2026). **Box 1** summarizes useful links for standards, best practices, and recommended software solutions.

Where should I submit my genome report?

Many journals, including *GBE*, are willing to accept manuscripts describing new genome assemblies, and your choice should be guided both by your specific desired target audience and by the amount of analysis and additional data you plan to include in your report. Some journals expect extensive new analysis and biological insight, whereas others

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explicitly disclaim any analysis or conclusions beyond a simple description of the assembly. A non-exclusive list of some common venues for genome reports is provided in **Table 1**.

How will my genome report be reviewed, and how should I review a genome report?

Like any manuscript, expect your genome report to undergo peer review. Many of the journals that accept genome reports have specific reviewing editors and guidelines. Although specific criteria vary from journal to journal, most of the journals accepting genome reports emphasize rigor and thoroughness of the methods and data, and not in-depth analysis or novel conclusions (in fact, the latter are specifically disallowed in many genome report formats; see **Table 1**).

With the ongoing rapid pace of genome sequencing, peer reviewers for genome reports are in high demand. Fortunately, genome reports are often quite straightforward to review and frequently require less time and effort than reviewing a more typical scientific paper. If asked to serve as a reviewer for a genome report, what should you look out for? Many of the journals accepting these reports have clear guidelines or even reviewer checklists that simplify the task. A comprehensive review of a genome note should focus on whether the authors have provided all of the necessary data, metadata, and metrics in a clear and accessible manner, with appropriate figures and tables. The reviewer should check that the methods used are well-supported by current standards and best practices (see above),

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and also that the methods are applicable to the organism in question. In all cases—especially in places where standards do not currently exist (Tvedte, et al. 2026)—the emphasis should be on whether all methods and protocols are thoroughly described and available to the point where results can be reproduced by other researchers. We recommend that a reviewer with subject matter expertise on the sequenced species be included to evaluate if prior work establishing karyotype and estimates of genome size and any biological particularities that might impact a sequencing project (e.g. haplodiploidy or otherwise unbalanced chromosome counts, sex chromosome system, repeat element or endosymbiont dynamics, genome silencing, or any other unusual chromosome biology) is accounted for. Unlike with more typical research papers, reviewers of genome notes should not generally be concerned with questions of significance, novelty (other than the fact that the relevant genome has not previously been sequenced to high quality), or biological insight. In **Supplemental File 2**, we provide a comprehensive checklist that includes and in many places exceeds the minimal criteria requested by most journals.

Looking ahead

It is remarkable that in the space of two decades, the once monumental achievement of producing a draft-quality eukaryotic genome has been replaced by almost routine generation of chromosome-level genome assemblies and annotations. While each new genome brings with it the potential to unlock new scientific insights, having a fully detailed

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and easily accessible genome report for each should also be a major goal. Such genome reports should be open, FAIR, and adhere to all current standards for metadata, methods, metrics, and data deposition. The journals in **Table 1** all take important steps toward this ideal. Looking ahead, we encourage these journals to adopt a common, machine-readable standard that will allow for easy summarization and re-use of genome data and metadata yet be flexible enough to incorporate ongoing improvements to standards and metrics. Genome reports can also be understood as a specialized form of "data paper" (Chavan and Penev 2011): a citable, human-readable metadata record for a genome assembly that links together sample provenance, methods, quality metrics, repository accessions, annotations, and conditions for reuse. Framing genome reports in this way helps to shift the emphasis away from journal-specific article formats and toward a more portable function: the genome report serves as a descriptive layer that helps make a genome assembly findable, accessible to humans and machines, interoperable across systems, and reusable across repositories and publication venues (i.e., the Genome Report as a FAIR data object). This perspective also strengthens the rationale for developing common, machine-readable reporting standards, since the report is not merely a notice of publication but part of the metadata infrastructure for the genome itself. To that end, an automated template that can create a suitable genome report would greatly facilitate the process for authors and reviewers alike; such a template could enforce inclusion of all relevant data, metadata, metrics, and methods descriptions, and could be customizable to meet journal-specific additional requirements. Moreover, although in Table 2 we list 22 recommended items for reporting in publications, only two journals—*G3* and *GBE*—require

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or suggest a majority of these. We therefore also encourage journals to re-evaluate their genome reports criteria with an eye toward ensuring that all necessary metadata, methods, metrics, and data are provided when genome assemblies are published. Genome sequence and annotation data are a fundamental resource in this genomic era, and comprehensive genome reports for each newly-sequenced genome are a critical component for ensuring that these essential data become and remain maximally accessible, accurate, and usable.

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presented in this paper are those of the author(s) and do not necessarily reflect the views of the NIH or the U.S. Department of Health and Human Services. MSH was supported by NIH grant U24 GM142435.

Data Availability Statement

The data underlying this article are available in the article and in its online supplementary material.

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Table 1: Journals with a "Genome Reports" article type*

Journal	Article type	Scope and Restrictions	Format	Special review criteria?	Special review process?	Citation/Link
Ecology&Evolution	Genetics Notes	genome assemblies and significant reassemblies	flexible	focus on rigor and validity of methods	yes	(Jenkins, et al. 2024)
F1000	Genome Notes	data analysis/conclusions not allowed	template	yes	yes	https://f1000research.com/for-authors/article-guidelines/genome-notes
G3	Genome Reports	new assemblies	specific	yes	yes	https://academic.oup.com/g3journal/pages/general-instructions
Genome Biol. Evol.	Genome Resources	draft genomes generally not allowed; extensive analysis not allowed	specific	yes	yes	https://academic.oup.com/gbe/pages/General_Instructions#Types%20of%20Manuscripts
GigaBytes	Research articles, Data notes	analysis/conclusions not allowed	suggested sections	yes	yes	personal communication from the editor
GigaScience	Research articles, Data notes	draft not allowed, T2T preferred; analysis/conclusions not allowed	suggested sections	yes	yes	personal communication from the editor
J. Insect Science	Molecular Entomological Genetics	insects only	flexible	no	no	https://academic.oup.com/jinsectscience/pages/Instructions_To_Authors#Article%20Types
micropublication.org			forthcoming	no	no	personal communication from the editor
MRA (Microbiology Resource Announcements)		microbes only	template	not described	not described	https://journals.asm.org/journal/mra/article-types#announcements

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Scientific Data	Data Descriptors	flexible	no	no	personal communication from the editor
Welcome Open Research	Data Note	template	no	not described	(Threlfall and Blaxter 2021)

*Journals surveyed in this Table were selected by the authors and are not intended as a comprehensive survey of biological sciences journals

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Table 2: Required/Recommended Genome Report information from 10 surveyed journals*

	Required	Recommended	Adoption frequency, % (DToL/Other)
Metadata			
description of sample source	6	-	100/100
ethics statement on sample collection	3	2	100/12
Other metadata	1	n/a	n/a
Methods			
General description of methods used	10	-	100/100
scripts/pipelines available	6	1	100/88
Metrics			
genome size estimation	1	2	100/82
assembly size	3	-	100/100
coverage estimation	2	-	100/74
contig N50	5	-	100/100
genome completeness (e.g. BUSCO/compleasm/OMark in genome mode)	4	-	100/100

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base accuracy (QV)	2	1	100/10
k-mer completeness (e.g. Merqury)	2	1	100/22
percent of contigs placed in chromosomes	1		100/54
Haplotype assembly/phasing	-	-	100/28
Sex determination identification/assembly	-	-	92/44
Organelle assembly	-	-	100/16
Genome analysis and Annotation			
Contamination screening	1	-	100/58
repeat and/or transposable element analysis	2	-	0/86
protein-coding annotation	4	-	44/94
BUSCO annotation in protein mode	-	1	0/56
Data			
submission of sequence/assembly data to an INSDC database	10	-	100/90
submission of annotation to an INSDC/other accessible database	1	2	44/72

*_microPublication Biology_ was included in our survey but is not counted among the 10 journals, as they have not yet formalized their Genome Report guidelines. See Table 1 for a list of journals. “n/a”, not applicable. Data for adoption frequency are taken from Tvedte et al.(Tvedte, et al. 2026). These numbers refer to an analysis of included data in a set of 100 publications of arthropod genomes, 50 published by the Darwin Tree of Life project as “genome notes” in Wellcome Open Research (“DtoL”) and 50 randomly selected from

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other journals. As only papers on arthropod genomes were included in that analysis, the numbers may show some bias toward community standards, but still point to wide adoption of certain reporting standards in at least some communities irrespective of explicit journal requirements.

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Box 1: Useful Resources for Genome Projects

Metadata resources:

NCBI BioSample attribute specifications:

https://www.ncbi.nlm.nih.gov/biosample/docs/packages/

NCBI BioSample home page:

https://www.ncbi.nlm.nih.gov/biosample/

EBI BioSamples home page:

https://www.ebi.ac.uk/biosamples/

FAIR BioHeaders specification and tools repository, which describes required metadata fields:

https://github.com/FAIR-bioHeaders

Minimum Information about any (x) Sequence (MIxS) checklist:
https://genomicsstandardsconsortium.github.io/mixs/

A comprehensive list of repositories for genome project data and metadata:
https://www.earthbiogenome.org/it-and-informatics-standards

Visualization resources:

BlobTk (conda, pip, prebuilt binaries, Docker, Singularity):

https://github.com/genomehubs/blobtk/wiki

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Standards and Tools/Software resources:

All Earth BioGenome subcommittee reports and recommendations:
https://www.earthbiogenome.org/reports-landing-page

Earth BioGenome Project assembly recommendations:
https://www.earthbiogenome.org/report-on-assembly-recommendations

Earth BioGenome assembly standards:
https://www.earthbiogenome.org/report-on-assembly-standards

Earth BioGenome Project recommended annotation tools:
https://www.earthbiogenome.org/report-on-annotation-recommended-tools

The Darwin Tree of Life's data processing pipelines:
https://pipelines.tol.sanger.ac.uk/pipelines

The Nagoya Protocol on Access and Benefit-sharing:
https://www.cbd.int/abs/default.shtml

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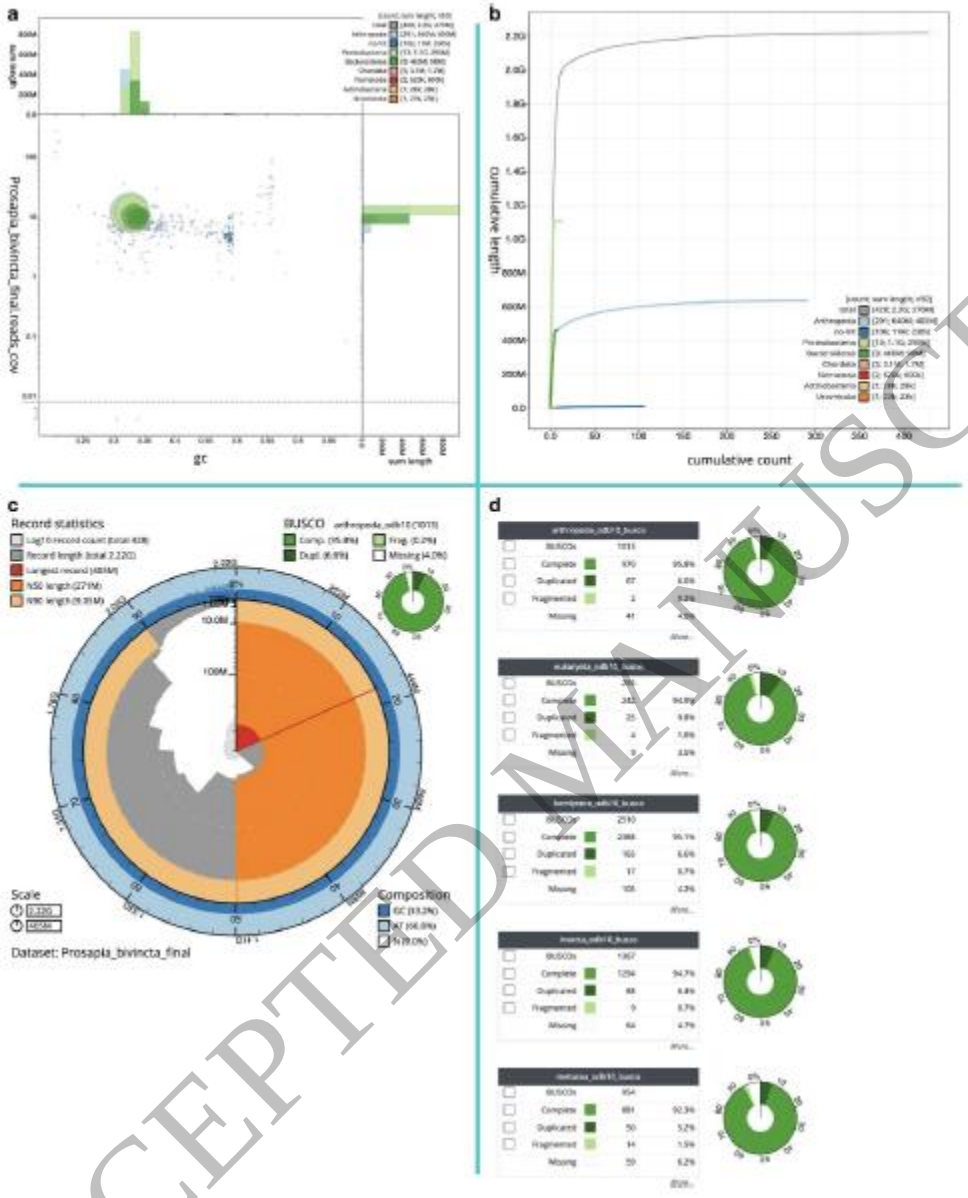


Figure 1: Example of a BlobTools report. The *Prosapia bicincta* genome was assembled using the "otb" pipeline following sequencing using a PacBio Sequel lie and generation of a Hi-C contact map (Molik, et al. 2024). BlobToolKit (Challis, et al. 2020) was used to generate the assembly report. (a) blobplot, showing contig length and GC content, (b) cumulative plots showing length of total contigs and their assignment of reads to orders, (c) snail plot showing record statistics and Busco, and (d) Busco plots showing complete buscos of several taxonomic classifications (while this report shows the genome assembly

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after contamination removal, likely misattributed contigs remain), ordered alphabetically.

Public domain image.

Alt text: Graphs of genome assembly metrics in BlobTools layout with subfigure labeled from a to d, each showing a different set of metrics.

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